

The 8 queen's problem is stated as: Place 8 queens on a chessboard such that no two queens attack each other. Considering this problem, complete the following tasks considering a valid heuristic function with justification.

Q1 Generate 50 random initial boards. Run steepest -ascent hill climbing on each and record initial heuristic value, final heuristic value, number of steps taken and status solved or fail. Prove the presence of a local minimum.

Q2 Repeat the above experiment on first choice, random restart and simulated annealing variants of hill climbing and compare your results.